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1. Find the exact value of the improper integral

$$\int_0^4 \frac{1}{\sqrt{x(4-x)}}\,\mathrm{d}x \qquad \textbf{[5]}$$

Solution

Let

$$I = \int_0^4 \frac{1}{\sqrt{x(4-x)}}\,\mathrm{d}x$$

This is an improper integral because the denominator is 0 at $x = 0$ and $x = 4$. So we write

$$I = \lim_{\substack{a\to 0^+\\ b\to 4^-}} \int_a^b \frac{1}{\sqrt{x(4-x)}}\,\mathrm{d}x$$

Now simplify the expression inside the square root:

$$x(4-x) = 4x - x^2 = 4 - (x^2 - 4x + 4) = 4 - (x-2)^2$$

So

$$I = \lim_{\substack{a\to 0^+\\ b\to 4^-}} \int_a^b \frac{1}{\sqrt{4-(x-2)^2}}\,\mathrm{d}x$$

Let

$$u = x - 2$$

Then

$$\mathrm{d}u = \mathrm{d}x$$

and the limits become

$$x = a \Rightarrow u = a - 2, \qquad x = b \Rightarrow u = b - 2$$

Hence

$$I = \lim_{\substack{a\to 0^+\\ b\to 4^-}} \int_{a-2}^{b-2} \frac{1}{\sqrt{4-u^2}}\,\mathrm{d}u$$

Using the standard result

$$\int \frac{1}{\sqrt{a^2-u^2}}\,\mathrm{d}u = \arcsin\left(\frac{u}{a}\right) + c$$

with $a = 2$, we get

$$I = \lim_{\substack{a\to 0^+\\ b\to 4^-}} \left[\arcsin\left(\frac{u}{2}\right)\right]_{a-2}^{b-2}$$

So

$$I = \lim_{\substack{a\to 0^+\\ b\to 4^-}} \left(\arcsin\left(\frac{b-2}{2}\right) - \arcsin\left(\frac{a-2}{2}\right)\right)$$

Now

$$\frac{b-2}{2} \to 1 \quad \text{as } b \to 4^-$$

and

$$\frac{a-2}{2} \to -1 \quad \text{as } a \to 0^+$$

Therefore

$$I = \arcsin(1) - \arcsin(-1) = \frac{\pi}{2} - \left(-\frac{\pi}{2}\right) = \pi$$

The exact value of the integral is π

2. Show that

$$\int_0^3 \frac{1}{x^2 - 2x + 5}\,\mathrm{d}x = \frac{1}{2}\arctan(3)$$ **[5]**

Solution

Let

$$I = \int_0^3 \frac{1}{x^2 - 2x + 5}\,\mathrm{d}x$$

First complete the square in the denominator:

$$x^2 - 2x + 5 = (x-1)^2 + 4 = (x-1)^2 + 2^2$$

So

$$I = \int_0^3 \frac{1}{(x-1)^2 + 2^2}\,\mathrm{d}x$$

Now use the standard result from the formula booklet:

$$\int \frac{1}{u^2 + a^2}\,\mathrm{d}u = \frac{1}{a}\arctan\left(\frac{u}{a}\right) + c$$

Here, take $u = x - 1$, so $\mathrm{d}u = \mathrm{d}x$, and $a = 2$. Hence

$$I = \frac{1}{2}\left[\arctan\left(\frac{x-1}{2}\right)\right]_0^3$$

Substituting the limits gives

$$\begin{aligned} I &= \frac{1}{2}\left(\arctan\left(\frac{3-1}{2}\right) - \arctan\left(\frac{0-1}{2}\right)\right) \\ &= \frac{1}{2}\left(\arctan(1) - \arctan\left(-\frac{1}{2}\right)\right) \\ &= \frac{1}{2}\left(\frac{\pi}{4} + \arctan\left(\frac{1}{2}\right)\right) \end{aligned}$$

To simplify this, let

$$\theta = \frac{\pi}{4} + \arctan\left(\frac{1}{2}\right)$$

Then

$$\begin{aligned} \tan\theta &= \frac{\tan\left(\frac{\pi}{4}\right) + \tan\left(\arctan\left(\frac{1}{2}\right)\right)}{1 - \tan\left(\frac{\pi}{4}\right)\tan\left(\arctan\left(\frac{1}{2}\right)\right)} \\ &= \frac{1 + \frac{1}{2}}{1 - 1 \cdot \frac{1}{2}} \\ &= \frac{\frac{3}{2}}{\frac{1}{2}} \\ &= 3 \end{aligned}$$

Also, both $\frac{\pi}{4}$ and $\arctan\left(\frac{1}{2}\right)$ are between 0 and $\frac{\pi}{2}$, so θ is in the first quadrant. Therefore

$$\theta = \arctan(3)$$

Hence

$$I = \frac{1}{2}\arctan(3)$$

So

$$\int_0^3 \frac{1}{x^2 - 2x + 5}\,\mathrm{d}x = \frac{1}{2}\arctan(3)$$

as required.

3. (i) Prove, from definitions involving exponentials, that

$$1 - \tanh^2 u = \operatorname{sech}^2 u$$ **[3]**

(ii) Prove that, for $|y| < 1$,

$$\operatorname{artanh} y = \frac{1}{2} \ln\left(\frac{1+y}{1-y}\right)$$ **[4]**

(iii) Use the substitution $x = 4\tanh u$ to show that, for $|x| < 4$,

$$\int \frac{1}{16 - x^2}\,\mathrm{d}x = \frac{1}{4}\operatorname{artanh}\frac{x}{4} + c$$

Hence deduce that

$$\int \frac{1}{16 - x^2}\,\mathrm{d}x = \frac{1}{8}\ln\left(\frac{4+x}{4-x}\right) + c$$

where c is an arbitrary constant. **[6]**

(iv) By first expressing $-t^2 + 6t + 7$ in completed square form, show that

$$\int_1^5 \frac{1}{-t^2 + 6t + 7}\,\mathrm{d}t = \frac{1}{4}\ln 3$$ **[5]**

Solution

(i) Using the exponential definitions,

$$\tanh u = \frac{e^u - e^{-u}}{e^u + e^{-u}} \qquad \text{and} \qquad \operatorname{sech} u = \frac{2}{e^u + e^{-u}}$$

Then

$$\begin{aligned}
1 - \tanh^2 u &= 1 - \left(\frac{e^u - e^{-u}}{e^u + e^{-u}}\right)^2 \\
&= \frac{(e^u + e^{-u})^2 - (e^u - e^{-u})^2}{(e^u + e^{-u})^2} \\
&= \frac{(e^{2u} + 2 + e^{-2u}) - (e^{2u} - 2 + e^{-2u})}{(e^u + e^{-u})^2} \\
&= \frac{4}{(e^u + e^{-u})^2} \\
&= \left(\frac{2}{e^u + e^{-u}}\right)^2 \\
&= \operatorname{sech}^2 u
\end{aligned}$$

Hence

$$1 - \tanh^2 u = \operatorname{sech}^2 u$$

(ii) Let

$$u = \operatorname{artanh} y$$

so that

$$y = \tanh u$$

Using the exponential form of $\tanh u$,

$$y = \frac{e^u - e^{-u}}{e^u + e^{-u}}$$

Multiply top and bottom by e^u:

$$y = \frac{e^{2u} - 1}{e^{2u} + 1}$$

Now solve for u:

$$\begin{aligned} y(e^{2u} + 1) &= e^{2u} - 1 \\ ye^{2u} + y &= e^{2u} - 1 \\ e^{2u} - ye^{2u} &= 1 + y \\ e^{2u}(1 - y) &= 1 + y \\ e^{2u} &= \frac{1+y}{1-y} \end{aligned}$$

Since $|y| < 1$, both $1 + y$ and $1 - y$ are positive, so taking logarithms is valid:

$$2u = \ln\left(\frac{1+y}{1-y}\right)$$

Therefore

$$u = \frac{1}{2}\ln\left(\frac{1+y}{1-y}\right)$$

But $u = \operatorname{artanh} y$, so

$$\operatorname{artanh} y = \frac{1}{2}\ln\left(\frac{1+y}{1-y}\right)$$

(iii) Let

$$x = 4\tanh u$$

Then

$$\frac{\mathrm{d}x}{\mathrm{d}u} = 4\operatorname{sech}^2 u \qquad \Rightarrow \qquad \mathrm{d}x = 4\operatorname{sech}^2 u\,\mathrm{d}u$$

Also,

$$\begin{aligned} 16 - x^2 &= 16 - 16\tanh^2 u \\ &= 16(1 - \tanh^2 u) \\ &= 16\operatorname{sech}^2 u \end{aligned}$$

using part (i).

So

$$\begin{aligned} \int \frac{1}{16 - x^2}\,\mathrm{d}x &= \int \frac{1}{16\operatorname{sech}^2 u}\left(4\operatorname{sech}^2 u\,\mathrm{d}u\right) \\ &= \frac{1}{4}\int 1\,\mathrm{d}u \\ &= \frac{u}{4} + c \end{aligned}$$

Now $x = 4\tanh u$ gives

$$\tanh u = \frac{x}{4}$$

so

$$u = \operatorname{artanh}\frac{x}{4}$$

Hence

$$\int \frac{1}{16 - x^2}\,\mathrm{d}x = \frac{1}{4}\operatorname{artanh}\frac{x}{4} + c$$

for $|x| < 4$.

Now use part (ii) with $y = \frac{x}{4}$. Since $|x| < 4$, we have $\left|\frac{x}{4}\right| < 1$, so

$$\operatorname{artanh}\frac{x}{4} = \frac{1}{2}\ln\left(\frac{1+\frac{x}{4}}{1-\frac{x}{4}}\right) = \frac{1}{2}\ln\left(\frac{4+x}{4-x}\right)$$

Therefore

$$\int \frac{1}{16-x^2}\,\mathrm{d}x = \frac{1}{4}\cdot\frac{1}{2}\ln\left(\frac{4+x}{4-x}\right)+c = \frac{1}{8}\ln\left(\frac{4+x}{4-x}\right)+c$$

So

$$\int \frac{1}{16-x^2}\,\mathrm{d}x = \frac{1}{8}\ln\left(\frac{4+x}{4-x}\right)+c$$

(iv) First complete the square:

$$\begin{aligned}-t^2+6t+7 &= -(t^2-6t+9)+16\\ &= 16-(t-3)^2\end{aligned}$$

So

$$\int_1^5 \frac{1}{-t^2+6t+7}\,\mathrm{d}t = \int_1^5 \frac{1}{16-(t-3)^2}\,\mathrm{d}t$$

Let

$$x = t-3$$

Then $\mathrm{d}x = \mathrm{d}t$, and the limits become

$$t=1 \Rightarrow x=-2, \qquad t=5 \Rightarrow x=2$$

Hence

$$\int_1^5 \frac{1}{-t^2+6t+7}\,\mathrm{d}t = \int_{-2}^2 \frac{1}{16-x^2}\,\mathrm{d}x$$

Using part (iii),

$$\int \frac{1}{16-x^2}\,\mathrm{d}x = \frac{1}{8}\ln\left(\frac{4+x}{4-x}\right)+c$$

Therefore

$$\begin{aligned}\int_{-2}^2 \frac{1}{16-x^2}\,\mathrm{d}x &= \left[\frac{1}{8}\ln\left(\frac{4+x}{4-x}\right)\right]_{-2}^2\\ &= \frac{1}{8}\ln\left(\frac{6}{2}\right)-\frac{1}{8}\ln\left(\frac{2}{6}\right)\\ &= \frac{1}{8}\ln 3-\frac{1}{8}\ln\left(\frac{1}{3}\right)\\ &= \frac{1}{8}\ln 3+\frac{1}{8}\ln 3\\ &= \frac{1}{4}\ln 3\end{aligned}$$

Hence

$$\int_1^5 \frac{1}{-t^2+6t+7}\,\mathrm{d}t = \frac{1}{4}\ln 3$$

4. (i) For $x > 5$, find

$$\int \frac{1}{\sqrt{x^2 - 6x + 5}}\,\mathrm{d}x$$

expressing your answer first in terms of arcosh and hence in logarithmic form. **[3]**

(ii) Use integration by substitution with $x - 3 = 2\cosh u$ to show that, for $x > 5$,

$$\int \sqrt{x^2 - 6x + 5}\,\mathrm{d}x = 2\left(\frac{x-3}{2}\sqrt{\frac{(x-3)^2}{4} - 1} - \ln\left|\frac{x-3}{2} + \sqrt{\frac{(x-3)^2}{4} - 1}\right|\right) + c$$

where c is an arbitrary constant. **[7]**

Solution

(i) First complete the square:

$$x^2 - 6x + 5 = (x-3)^2 - 4$$

Since $x > 5$, we have $\dfrac{x-3}{2} > 1$, so we may let

$$x - 3 = 2\cosh u \qquad (u > 0)$$

Then

$$\frac{\mathrm{d}x}{\mathrm{d}u} = 2\sinh u \qquad \Rightarrow \qquad \mathrm{d}x = 2\sinh u\,\mathrm{d}u$$

and

$$\begin{aligned}\sqrt{x^2 - 6x + 5} &= \sqrt{(x-3)^2 - 4}\\ &= \sqrt{4\cosh^2 u - 4}\\ &= 2\sqrt{\cosh^2 u - 1}\\ &= 2\sinh u\end{aligned}$$

since $u > 0$.

So

$$\begin{aligned}\int \frac{1}{\sqrt{x^2 - 6x + 5}}\,\mathrm{d}x &= \int \frac{1}{2\sinh u}\,(2\sinh u\,\mathrm{d}u)\\ &= \int 1\,\mathrm{d}u\\ &= u + c\end{aligned}$$

Now

$$\cosh u = \frac{x-3}{2}$$

so

$$u = \operatorname{arcosh}\left(\frac{x-3}{2}\right)$$

Hence

$$\int \frac{1}{\sqrt{x^2 - 6x + 5}}\,\mathrm{d}x = \operatorname{arcosh}\left(\frac{x-3}{2}\right) + c$$

Using the standard result

$$\operatorname{arcosh} z = \ln\left(z + \sqrt{z^2 - 1}\right)$$

we get

$$\int \frac{1}{\sqrt{x^2 - 6x + 5}}\,\mathrm{d}x = \ln\left(\frac{x-3}{2} + \sqrt{\frac{(x-3)^2}{4} - 1}\right) + c$$

So the answer is

$$\operatorname{arcosh}\left(\frac{x-3}{2}\right)+c$$

and hence

$$\ln\left(\frac{x-3}{2}+\sqrt{\frac{(x-3)^2}{4}-1}\right)+c$$

(ii) Let

$$I=\int\sqrt{x^2-6x+5}\,\mathrm{d}x$$

and use the substitution

$$x-3=2\cosh u \qquad (x>5)$$

Then, as above,

$$\sqrt{x^2-6x+5}=2\sinh u \qquad \text{and} \qquad \mathrm{d}x=2\sinh u\,\mathrm{d}u$$

So

$$\begin{aligned} I &= \int(2\sinh u)(2\sinh u)\,\mathrm{d}u \\ &= 4\int\sinh^2 u\,\mathrm{d}u \end{aligned}$$

Use the identity

$$\cosh 2u=2\sinh^2 u+1$$

so that

$$2\sinh^2 u=\cosh 2u-1$$

Therefore

$$\begin{aligned} I &= 2\int(\cosh 2u-1)\,\mathrm{d}u \\ &= 2\left(\frac{1}{2}\sinh 2u-u\right)+c \\ &= \sinh 2u-2u+c \end{aligned}$$

Now express this in terms of x.
Since

$$\cosh u=\frac{x-3}{2}$$

we have

$$\sinh u=\sqrt{\cosh^2 u-1}=\sqrt{\frac{(x-3)^2}{4}-1}$$

because $u>0$.
Also

$$\begin{aligned} \sinh 2u &= 2\sinh u\cosh u \\ &= 2\left(\sqrt{\frac{(x-3)^2}{4}-1}\right)\left(\frac{x-3}{2}\right) \\ &= (x-3)\sqrt{\frac{(x-3)^2}{4}-1} \end{aligned}$$

Hence

$$\begin{aligned} I &= (x-3)\sqrt{\frac{(x-3)^2}{4}-1}-2u+c \\ &= 2\left(\frac{x-3}{2}\sqrt{\frac{(x-3)^2}{4}-1}-u\right)+c \end{aligned}$$

From part (ii),

$$u = \operatorname{arcosh}\left(\frac{x-3}{2}\right) = \ln\left|\frac{x-3}{2} + \sqrt{\frac{(x-3)^2}{4} - 1}\right|$$

Therefore

$$\int \sqrt{x^2 - 6x + 5}\,\mathrm{d}x = 2\left(\frac{x-3}{2}\sqrt{\frac{(x-3)^2}{4} - 1} - \ln\left|\frac{x-3}{2} + \sqrt{\frac{(x-3)^2}{4} - 1}\right|\right) + c$$

This is the required result.

5.

$$g(x) = \frac{2x^2 - 3x + 31}{x^3 - 3x^2 + 9x + 13}$$

(a) Express $g(x)$ in the form

$$\frac{P}{x+1} + \frac{Q}{x^2 - 4x + 13}$$

where P and Q are constants to be found. **[3]**

(b) Find $\int g(x)\,\mathrm{d}x$, giving your answer in the form

$$A \ln|x+1| + B \arctan\left(\frac{x-2}{3}\right) + c$$

where A and B are constants to be found. **[4]**

(c) Hence show that $\int_0^\infty g(x)\,\mathrm{d}x$ diverges. **[2]**

Solution

(a) First factorise the denominator:

$$x^3 - 3x^2 + 9x + 13 = (x+1)(x^2 - 4x + 13)$$

So

$$g(x) = \frac{2x^2 - 3x + 31}{(x+1)(x^2 - 4x + 13)}$$

and we write

$$\frac{2x^2 - 3x + 31}{(x+1)(x^2 - 4x + 13)} = \frac{P}{x+1} + \frac{Q}{x^2 - 4x + 13}$$

Multiplying through by $(x+1)(x^2 - 4x + 13)$ gives

$$2x^2 - 3x + 31 = P(x^2 - 4x + 13) + Q(x+1)$$

Expand the right-hand side:

$$\begin{aligned} 2x^2 - 3x + 31 &= Px^2 - 4Px + 13P + Qx + Q \\ &= Px^2 + (-4P + Q)x + (13P + Q) \end{aligned}$$

Now compare coefficients with

$$2x^2 - 3x + 31$$

So

$$P = 2$$

and

$$-4P + Q = -3$$

Substituting $P = 2$:

$$\begin{aligned} -8 + Q &= -3 \\ Q &= 5 \end{aligned}$$

Checking the constant term:

$$13P + Q = 13(2) + 5 = 31$$

which is correct.

Hence

$$P = 2, \qquad Q = 5$$

(b) From part (a),

$$g(x) = \frac{2}{x+1} + \frac{5}{x^2 - 4x + 13}$$

Complete the square in the quadratic:

$$x^2 - 4x + 13 = (x-2)^2 + 9$$

So

$$g(x) = \frac{2}{x+1} + \frac{5}{(x-2)^2 + 9}$$

Therefore

$$\int g(x)\,\mathrm{d}x = \int \frac{2}{x+1}\,\mathrm{d}x + \int \frac{5}{(x-2)^2+9}\,\mathrm{d}x$$

For the first integral,

$$\int \frac{2}{x+1}\,\mathrm{d}x = 2\ln|x+1|$$

For the second integral, use the standard result

$$\int \frac{1}{u^2+a^2}\,\mathrm{d}u = \frac{1}{a}\arctan\left(\frac{u}{a}\right) + c$$

Here $u = x - 2$ and $a = 3$, so

$$\int \frac{5}{(x-2)^2+9}\,\mathrm{d}x = 5\int \frac{1}{(x-2)^2+3^2}\,\mathrm{d}x = \frac{5}{3}\arctan\left(\frac{x-2}{3}\right)$$

Hence

$$\int g(x)\,\mathrm{d}x = 2\ln|x+1| + \frac{5}{3}\arctan\left(\frac{x-2}{3}\right) + c$$

So

$$A = 2, \qquad B = \frac{5}{3}$$

(c) To consider the improper integral, write

$$\int_0^\infty g(x)\,\mathrm{d}x = \lim_{t\to\infty}\int_0^t g(x)\,\mathrm{d}x$$

Using the antiderivative from part (b),

$$\begin{aligned}
\int_0^t g(x)\,\mathrm{d}x &= \left[2\ln|x+1| + \frac{5}{3}\arctan\left(\frac{x-2}{3}\right)\right]_0^t \\
&= 2\ln(t+1) + \frac{5}{3}\arctan\left(\frac{t-2}{3}\right) - \left(2\ln 1 + \frac{5}{3}\arctan\left(-\frac{2}{3}\right)\right) \\
&= 2\ln(t+1) + \frac{5}{3}\arctan\left(\frac{t-2}{3}\right) + \frac{5}{3}\arctan\left(\frac{2}{3}\right)
\end{aligned}$$

Now as $t \to \infty$,

$$\arctan\left(\frac{t-2}{3}\right) \to \frac{\pi}{2}$$

so that term stays finite, but

$$2\ln(t+1) \to \infty$$

Therefore

$$\int_0^\infty g(x)\,\mathrm{d}x$$

diverges, in fact it diverges to $+\infty$.

6. Show that

$$\int_1^4 \frac{1}{\sqrt{10+8x-2x^2}}\,\mathrm{d}x = \frac{1}{\sqrt{2}}\left(\arcsin\left(\frac{2}{3}\right)+\arcsin\left(\frac{1}{3}\right)\right)$$ **[6]**

Solution

Let

$$I = \int_1^4 \frac{1}{\sqrt{10+8x-2x^2}}\,\mathrm{d}x$$

First complete the square in the expression under the root:

$$\begin{aligned} 10+8x-2x^2 &= -2(x^2-4x-5) \\ &= -2\big((x-2)^2-9\big) \\ &= 2\big(9-(x-2)^2\big) \end{aligned}$$

So

$$I = \int_1^4 \frac{1}{\sqrt{2\big(9-(x-2)^2\big)}}\,\mathrm{d}x = \frac{1}{\sqrt{2}}\int_1^4 \frac{1}{\sqrt{9-(x-2)^2}}\,\mathrm{d}x$$

Now reduce this to the standard form

$$\int \frac{1}{\sqrt{1-u^2}}\,\mathrm{d}u = \arcsin(u)+c$$

Let

$$u = \frac{x-2}{3}$$

Then

$$x-2 = 3u, \qquad \mathrm{d}x = 3\,\mathrm{d}u$$

Also, when $x=1$,

$$u = \frac{1-2}{3} = -\frac{1}{3}$$

and when $x=4$,

$$u = \frac{4-2}{3} = \frac{2}{3}$$

Substituting into the integral:

$$\begin{aligned} I &= \frac{1}{\sqrt{2}}\int_1^4 \frac{1}{\sqrt{9-(x-2)^2}}\,\mathrm{d}x \\ &= \frac{1}{\sqrt{2}}\int_{-1/3}^{2/3} \frac{3}{\sqrt{9-(3u)^2}}\,\mathrm{d}u \\ &= \frac{1}{\sqrt{2}}\int_{-1/3}^{2/3} \frac{3}{\sqrt{9-9u^2}}\,\mathrm{d}u \\ &= \frac{1}{\sqrt{2}}\int_{-1/3}^{2/3} \frac{3}{3\sqrt{1-u^2}}\,\mathrm{d}u \\ &= \frac{1}{\sqrt{2}}\int_{-1/3}^{2/3} \frac{1}{\sqrt{1-u^2}}\,\mathrm{d}u \end{aligned}$$

Using the standard result:

$$\begin{aligned} I &= \frac{1}{\sqrt{2}}\left[\arcsin(u)\right]_{-1/3}^{2/3} \\ &= \frac{1}{\sqrt{2}}\left(\arcsin\left(\frac{2}{3}\right)-\arcsin\left(-\frac{1}{3}\right)\right) \end{aligned}$$

Since $\arcsin(-a) = -\arcsin(a)$,

$$I = \frac{1}{\sqrt{2}}\left(\arcsin\left(\frac{2}{3}\right)+\arcsin\left(\frac{1}{3}\right)\right)$$

Hence

$$\int_1^4 \frac{1}{\sqrt{10+8x-2x^2}}\,\mathrm{d}x = \frac{1}{\sqrt{2}}\left(\arcsin\left(\frac{2}{3}\right)+\arcsin\left(\frac{1}{3}\right)\right)$$

7. **(a)** Explain why

$$\int_0^\infty \frac{1}{x^2 - 6x + 18}\,\mathrm{d}x$$

is an improper integral. **[1]**

(b) Show that

$$\int_0^\infty \frac{1}{x^2 - 6x + 18}\,\mathrm{d}x = k\pi$$

where k is a constant to be determined. **[4]**

Solution

(a) The integral is improper because the interval of integration is unbounded: the upper limit is ∞. So it is defined as

$$\int_0^\infty \frac{1}{x^2 - 6x + 18}\,\mathrm{d}x = \lim_{b\to\infty}\int_0^b \frac{1}{x^2 - 6x + 18}\,\mathrm{d}x$$

(b) Let

$$I = \int_0^\infty \frac{1}{x^2 - 6x + 18}\,\mathrm{d}x$$

First complete the square in the denominator:

$$x^2 - 6x + 18 = (x-3)^2 + 9 = (x-3)^2 + 3^2$$

Hence

$$I = \int_0^\infty \frac{1}{(x-3)^2 + 3^2}\,\mathrm{d}x$$

Since this is an improper integral, write it as a limit:

$$I = \lim_{b\to\infty}\int_0^b \frac{1}{(x-3)^2 + 3^2}\,\mathrm{d}x$$

Using the standard result

$$\int \frac{1}{u^2 + a^2}\,\mathrm{d}u = \frac{1}{a}\arctan\left(\frac{u}{a}\right) + c$$

with $u = x - 3$ and $a = 3$, we get

$$\int \frac{1}{(x-3)^2 + 3^2}\,\mathrm{d}x = \frac{1}{3}\arctan\left(\frac{x-3}{3}\right) + c$$

Therefore

$$\begin{aligned} I &= \lim_{b\to\infty}\left[\frac{1}{3}\arctan\left(\frac{x-3}{3}\right)\right]_0^b \\ &= \frac{1}{3}\lim_{b\to\infty}\left(\arctan\left(\frac{b-3}{3}\right) - \arctan(-1)\right) \end{aligned}$$

Now

$$\arctan\left(\frac{b-3}{3}\right) \to \frac{\pi}{2} \quad \text{as } b \to \infty$$

and

$$\arctan(-1) = -\frac{\pi}{4}$$

So

$$\begin{aligned} I &= \frac{1}{3}\left(\frac{\pi}{2} - \left(-\frac{\pi}{4}\right)\right) \\ &= \frac{1}{3}\left(\frac{3\pi}{4}\right) \\ &= \frac{\pi}{4} \end{aligned}$$

Hence

$$\int_0^\infty \frac{1}{x^2 - 6x + 18}\,\mathrm{d}x = \frac{\pi}{4} = k\pi$$

so

$$k = \frac{1}{4}$$

8. Evaluate

$$\int_0^4 \frac{1}{x^2 - 4x + 8}\,\mathrm{d}x$$

giving your answer as an exact multiple of π. **[8]**

Solution

First complete the square in the denominator:

$$x^2 - 4x + 8 = (x^2 - 4x + 4) + 4 = (x-2)^2 + 4 = (x-2)^2 + 2^2$$

So the integral becomes

$$\int_0^4 \frac{1}{x^2 - 4x + 8}\,\mathrm{d}x = \int_0^4 \frac{1}{(x-2)^2 + 2^2}\,\mathrm{d}x$$

Now use the standard result from the formula booklet:

$$\int \frac{1}{u^2 + a^2}\,\mathrm{d}u = \frac{1}{a}\arctan\left(\frac{u}{a}\right) + c$$

Here we can take

$$u = x - 2 \qquad \text{so} \qquad \mathrm{d}u = \mathrm{d}x$$

Therefore

$$\int_0^4 \frac{1}{(x-2)^2 + 2^2}\,\mathrm{d}x = \left[\frac{1}{2}\arctan\left(\frac{x-2}{2}\right)\right]_0^4$$

Now substitute the limits:

$$\begin{aligned}
\left[\frac{1}{2}\arctan\left(\frac{x-2}{2}\right)\right]_0^4 &= \frac{1}{2}\arctan\left(\frac{4-2}{2}\right) - \frac{1}{2}\arctan\left(\frac{0-2}{2}\right) \\
&= \frac{1}{2}\arctan(1) - \frac{1}{2}\arctan(-1) \\
&= \frac{1}{2}\cdot\frac{\pi}{4} - \frac{1}{2}\cdot\left(-\frac{\pi}{4}\right) \\
&= \frac{1}{2}\left(\frac{\pi}{4} + \frac{\pi}{4}\right) \\
&= \frac{1}{2}\cdot\frac{\pi}{2} \\
&= \frac{\pi}{4}
\end{aligned}$$

Hence

$$\int_0^4 \frac{1}{x^2 - 4x + 8}\,\mathrm{d}x = \frac{\pi}{4}$$

So the exact value, as a multiple of π, is $\frac{\pi}{4}$.

9.

$$f(x) = \frac{6x+1}{4x^2-8x+13}$$

(a) Find $\int f(x)\,\mathrm{d}x$, giving your answer in the form

$$A\ln(4x^2-8x+13) + B\arctan\left(\frac{2x-2}{3}\right) + c$$

where c is an arbitrary constant and A and B are constants to be found. **[4]**

(b) Hence show that the mean value of $f(x)$ over the interval $\left[1, \frac{5}{2}\right]$ is

$$\frac{1}{2}\ln 2 + \frac{7\pi}{36}$$ **[3]**

(c) Hence write down the mean value of $3 - 2f(x)$ over the interval $\left[1, \frac{5}{2}\right]$. **[1]**

Solution

(a) Let

$$I = \int \frac{6x+1}{4x^2-8x+13}\,\mathrm{d}x$$

Write the numerator in terms of the derivative of the denominator:

$$6x+1 = \frac{3}{4}(8x-8) + 7$$

So

$$I = \frac{3}{4}\int \frac{8x-8}{4x^2-8x+13}\,\mathrm{d}x + 7\int \frac{1}{4x^2-8x+13}\,\mathrm{d}x$$

For the first integral,

$$\int \frac{g'(x)}{g(x)}\,\mathrm{d}x = \ln|g(x)| + c$$

and here

$$4x^2-8x+13 = (2x-2)^2+9 > 0$$

so no modulus is needed. Hence

$$\frac{3}{4}\int \frac{8x-8}{4x^2-8x+13}\,\mathrm{d}x = \frac{3}{4}\ln(4x^2-8x+13)$$

For the second integral, complete the square:

$$4x^2-8x+13 = (2x-2)^2+9$$

Let $u = 2x-2$, so $\mathrm{d}u = 2\,\mathrm{d}x$ and $\mathrm{d}x = \frac{1}{2}\,\mathrm{d}u$. Then

$$\begin{aligned} 7\int \frac{1}{(2x-2)^2+9}\,\mathrm{d}x &= 7\int \frac{1}{u^2+9}\cdot\frac{1}{2}\,\mathrm{d}u \\ &= \frac{7}{2}\int \frac{1}{u^2+9}\,\mathrm{d}u \end{aligned}$$

Using the standard result

$$\int \frac{1}{a^2+u^2}\,\mathrm{d}u = \frac{1}{a}\arctan\left(\frac{u}{a}\right) + c$$

with $a = 3$,

$$\begin{aligned}\frac{7}{2}\int \frac{1}{u^2+9}\,\mathrm{d}u &= \frac{7}{2}\cdot\frac{1}{3}\arctan\left(\frac{u}{3}\right)\\ &= \frac{7}{6}\arctan\left(\frac{u}{3}\right)\end{aligned}$$

Substituting back $u = 2x - 2$,

$$\int f(x)\,\mathrm{d}x = \frac{3}{4}\ln(4x^2 - 8x + 13) + \frac{7}{6}\arctan\left(\frac{2x-2}{3}\right) + c$$

Therefore

$$A = \frac{3}{4}, \qquad B = \frac{7}{6}$$

(b) From part (a), an antiderivative is

$$F(x) = \frac{3}{4}\ln(4x^2 - 8x + 13) + \frac{7}{6}\arctan\left(\frac{2x-2}{3}\right)$$

The mean value of $f(x)$ over $\left[1, \frac{5}{2}\right]$ is

$$\frac{1}{\frac{5}{2}-1}\int_1^{5/2} f(x)\,\mathrm{d}x = \frac{2}{3}\big(F(5/2) - F(1)\big)$$

Now

$$\begin{aligned}F\left(\frac{5}{2}\right) &= \frac{3}{4}\ln\left(4\cdot\frac{25}{4} - 8\cdot\frac{5}{2} + 13\right) + \frac{7}{6}\arctan\left(\frac{2\cdot\frac{5}{2}-2}{3}\right)\\ &= \frac{3}{4}\ln(18) + \frac{7}{6}\arctan(1)\\ &= \frac{3}{4}\ln(18) + \frac{7\pi}{24}\end{aligned}$$

Also

$$\begin{aligned}F(1) &= \frac{3}{4}\ln(4 - 8 + 13) + \frac{7}{6}\arctan(0)\\ &= \frac{3}{4}\ln(9)\end{aligned}$$

So

$$\begin{aligned}\frac{2}{3}\big(F(5/2) - F(1)\big) &= \frac{2}{3}\left(\frac{3}{4}\ln\frac{18}{9} + \frac{7\pi}{24}\right)\\ &= \frac{2}{3}\left(\frac{3}{4}\ln 2 + \frac{7\pi}{24}\right)\\ &= \frac{1}{2}\ln 2 + \frac{7\pi}{36}\end{aligned}$$

Hence the mean value is

$$\frac{1}{2}\ln 2 + \frac{7\pi}{36}$$

(c) Using linearity of the mean value,

$$\text{mean value of } (3 - 2f(x)) = 3 - 2 \times \text{mean value of } f(x)$$

So

$$\begin{aligned}\text{mean value of } (3 - 2f(x)) &= 3 - 2\left(\frac{1}{2}\ln 2 + \frac{7\pi}{36}\right)\\ &= 3 - \ln 2 - \frac{7\pi}{18}\end{aligned}$$

Therefore the mean value is

$$3 - \ln 2 - \frac{7\pi}{18}$$

10.

$$g(x) = \frac{4x-1}{x^2-4x+8}$$

Show that

$$\int g(x)\,\mathrm{d}x = A\arctan\left(\frac{x-2}{2}\right) + B\ln(x^2-4x+8) + c$$

where c is an arbitrary constant and A and B are constants to be found. **[5]**

Solution

First complete the square in the denominator:

$$x^2-4x+8 = (x-2)^2+4$$

Now rewrite the numerator so that part of it is the derivative of the denominator:

$$4x-1 = 2(2x-4)+7$$

So

$$\begin{aligned} g(x) &= \frac{4x-1}{x^2-4x+8} \\ &= \frac{2(2x-4)+7}{x^2-4x+8} \\ &= 2\frac{2x-4}{x^2-4x+8} + \frac{7}{(x-2)^2+4} \end{aligned}$$

Hence

$$\int g(x)\,\mathrm{d}x = 2\int \frac{2x-4}{x^2-4x+8}\,\mathrm{d}x + 7\int \frac{1}{(x-2)^2+4}\,\mathrm{d}x$$

For the first integral, let

$$u = x^2-4x+8 \qquad \text{so} \qquad \frac{\mathrm{d}u}{\mathrm{d}x} = 2x-4$$

Then

$$2\int \frac{2x-4}{x^2-4x+8}\,\mathrm{d}x = 2\int \frac{1}{u}\,\mathrm{d}u = 2\ln u = 2\ln(x^2-4x+8)$$

For the second integral, use the standard result

$$\int \frac{1}{a^2+u^2}\,\mathrm{d}u = \frac{1}{a}\arctan\left(\frac{u}{a}\right) + c$$

Here $u = x-2$ and $a = 2$, so

$$7\int \frac{1}{(x-2)^2+4}\,\mathrm{d}x = 7\cdot\frac{1}{2}\arctan\left(\frac{x-2}{2}\right)$$

Therefore

$$\int g(x)\,\mathrm{d}x = 2\ln(x^2-4x+8) + \frac{7}{2}\arctan\left(\frac{x-2}{2}\right) + c$$

This is of the form

$$\int g(x)\,\mathrm{d}x = A\arctan\left(\frac{x-2}{2}\right) + B\ln(x^2-4x+8) + c$$

with

$$A = \frac{7}{2}, \qquad B = 2$$

So the required constants are $A = \frac{7}{2}$ and $B = 2$.

11. You are given that $y = \operatorname{arcosh} x$, where $x \geq 1$.

(i) Show that

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{\sqrt{x^2 - 1}}$$ **[4]**

(ii) Show, by integrating the result in part (i), that

$$y = \ln\left(x + \sqrt{x^2 - 1}\right)$$ **[4]**

(iii) Hence show that

$$\int_{15/8}^{5/2} \frac{1}{\sqrt{4x^2 - 9}}\,\mathrm{d}x = \frac{1}{2}\left(\operatorname{arcosh}\frac{5}{3} - \operatorname{arcosh}\frac{5}{4}\right)$$

Express this answer in logarithmic form. **[4]**

Solution

(i) Since $y = \operatorname{arcosh} x$, we have

$$\cosh y = x$$

Differentiate implicitly with respect to x:

$$\sinh y \frac{\mathrm{d}y}{\mathrm{d}x} = 1$$

So

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{\sinh y}$$

Using the identity

$$\cosh^2 y - \sinh^2 y = 1$$

gives

$$\sinh^2 y = \cosh^2 y - 1 = x^2 - 1$$

Also $y = \operatorname{arcosh} x$ with $x \geq 1$, so $y \geq 0$, hence $\sinh y \geq 0$. Therefore

$$\sinh y = \sqrt{x^2 - 1}$$

and so

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{\sqrt{x^2 - 1}}$$

as required.

(ii) Integrating the result from part (ii),

$$y = \int \frac{1}{\sqrt{x^2 - 1}}\,\mathrm{d}x$$

Use the standard result

$$\int \frac{1}{\sqrt{x^2 - a^2}}\,\mathrm{d}x = \ln\left|x + \sqrt{x^2 - a^2}\right| + C$$

With $a = 1$,

$$y = \ln\left|x + \sqrt{x^2 - 1}\right| + C$$

Since $x \geq 1$, the quantity $x + \sqrt{x^2 - 1}$ is positive, so

$$y = \ln\left(x + \sqrt{x^2 - 1}\right) + C$$

Now when $x = 1$,

$$y = \operatorname{arcosh} 1 = 0$$

Hence

$$0 = \ln\left(1 + \sqrt{1^2 - 1}\right) + C = \ln 1 + C = C$$

So $C = 0$, and therefore

$$y = \ln\left(x + \sqrt{x^2 - 1}\right)$$

Alternative method (without quoting the standard result):

Let

$$I = \int \frac{1}{\sqrt{x^2 - 1}}\,\mathrm{d}x$$

Multiply the integrand by

$$\frac{x + \sqrt{x^2 - 1}}{x + \sqrt{x^2 - 1}}$$

to obtain

$$\begin{aligned} I &= \int \frac{x + \sqrt{x^2 - 1}}{\sqrt{x^2 - 1}\left(x + \sqrt{x^2 - 1}\right)}\,\mathrm{d}x \\ &= \int \frac{1 + \dfrac{x}{\sqrt{x^2 - 1}}}{x + \sqrt{x^2 - 1}}\,\mathrm{d}x \end{aligned}$$

Now

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(x + \sqrt{x^2 - 1}\right) = 1 + \frac{x}{\sqrt{x^2 - 1}}$$

so

$$I = \int \frac{f'(x)}{f(x)}\,\mathrm{d}x \quad \text{with} \quad f(x) = x + \sqrt{x^2 - 1}$$

Hence

$$I = \ln\left(x + \sqrt{x^2 - 1}\right) + C$$

since $x \geq 1$ implies $x + \sqrt{x^2 - 1} > 0$

Therefore

$$y = \ln\left(x + \sqrt{x^2 - 1}\right) + C$$

Now when $x = 1$,

$$y = \operatorname{arcosh} 1 = 0$$

so

$$0 = \ln\left(1 + \sqrt{1^2 - 1}\right) + C = \ln 1 + C = C$$

Thus $C = 0$, giving

$$y = \ln\left(x + \sqrt{x^2 - 1}\right)$$

(iii) Let

$$u = \frac{2x}{3}$$

Then

$$\mathrm{d}x = \frac{3}{2}\,\mathrm{d}u$$

and

$$\sqrt{4x^2 - 9} = \sqrt{9u^2 - 9} = 3\sqrt{u^2 - 1}$$

Also the limits become

$$x = \frac{15}{8} \Rightarrow u = \frac{5}{4}, \qquad x = \frac{5}{2} \Rightarrow u = \frac{5}{3}$$

So

$$\int_{15/8}^{5/2} \frac{1}{\sqrt{4x^2-9}}\,\mathrm{d}x = \int_{5/4}^{5/3} \frac{1}{3\sqrt{u^2-1}} \cdot \frac{3}{2}\,\mathrm{d}u$$
$$= \frac{1}{2}\int_{5/4}^{5/3} \frac{1}{\sqrt{u^2-1}}\,\mathrm{d}u$$

From part (ii), $\frac{\mathrm{d}}{\mathrm{d}u}(\operatorname{arcosh} u) = \frac{1}{\sqrt{u^2-1}}$, so

$$\int_{15/8}^{5/2} \frac{1}{\sqrt{4x^2-9}}\,\mathrm{d}x = \frac{1}{2}\left[\operatorname{arcosh} u\right]_{5/4}^{5/3} = \frac{1}{2}\left(\operatorname{arcosh}\frac{5}{3} - \operatorname{arcosh}\frac{5}{4}\right)$$

which is the required result.

Now express this in logarithmic form using part (iii):

$$\operatorname{arcosh} t = \ln\left(t + \sqrt{t^2-1}\right)$$

Hence

$$\operatorname{arcosh}\frac{5}{3} = \ln\left(\frac{5}{3} + \sqrt{\frac{25}{9}-1}\right) = \ln\left(\frac{5}{3}+\frac{4}{3}\right) = \ln 3$$

and

$$\operatorname{arcosh}\frac{5}{4} = \ln\left(\frac{5}{4} + \sqrt{\frac{25}{16}-1}\right) = \ln\left(\frac{5}{4}+\frac{3}{4}\right) = \ln 2$$

Therefore

$$\int_{15/8}^{5/2} \frac{1}{\sqrt{4x^2-9}}\,\mathrm{d}x = \frac{1}{2}(\ln 3 - \ln 2) = \frac{1}{2}\ln\left(\frac{3}{2}\right)$$

So the logarithmic form is

$$\frac{1}{2}\ln\left(\frac{3}{2}\right)$$

12. Determine each of the following integrals.

(i)

$$\int \frac{1}{9x^2 + 12x + 20}\,\mathrm{d}x \qquad \textbf{[4]}$$

(ii)

$$\int \frac{1}{\sqrt{45 + 12x - 9x^2}}\,\mathrm{d}x \qquad \textbf{[4]}$$

Solution

(i) Complete the square in the denominator:

$$\begin{aligned}
9x^2 + 12x + 20 &= 9\left(x^2 + \frac{4}{3}x\right) + 20 \\
&= 9\left[\left(x + \frac{2}{3}\right)^2 - \frac{4}{9}\right] + 20 \\
&= 9\left(x + \frac{2}{3}\right)^2 - 4 + 20 \\
&= 9\left(x + \frac{2}{3}\right)^2 + 16 \\
&= 9\left[\left(x + \frac{2}{3}\right)^2 + \left(\frac{4}{3}\right)^2\right]
\end{aligned}$$

So

$$\int \frac{1}{9x^2 + 12x + 20}\,\mathrm{d}x = \frac{1}{9}\int \frac{1}{\left(x + \frac{2}{3}\right)^2 + \left(\frac{4}{3}\right)^2}\,\mathrm{d}x$$

Let

$$u = x + \frac{2}{3}$$

so that $\mathrm{d}u = \mathrm{d}x$. Then

$$\int \frac{1}{9x^2 + 12x + 20}\,\mathrm{d}x = \frac{1}{9}\int \frac{1}{u^2 + \left(\frac{4}{3}\right)^2}\,\mathrm{d}u$$

Using the standard result

$$\int \frac{1}{x^2 + a^2}\,\mathrm{d}x = \frac{1}{a}\arctan\left(\frac{x}{a}\right) + C$$

with $a = \frac{4}{3}$,

$$\begin{aligned}
\int \frac{1}{9x^2 + 12x + 20}\,\mathrm{d}x &= \frac{1}{9}\cdot\frac{1}{4/3}\arctan\left(\frac{u}{4/3}\right) + C \\
&= \frac{1}{12}\arctan\left(\frac{3u}{4}\right) + C \\
&= \frac{1}{12}\arctan\left(\frac{3x + 2}{4}\right) + C
\end{aligned}$$

Hence,

$$\int \frac{1}{9x^2 + 12x + 20}\,\mathrm{d}x = \frac{1}{12}\arctan\left(\frac{3x + 2}{4}\right) + C$$

(ii) Rewrite the expression under the root in completed-square form:

$$\begin{aligned}
45 + 12x - 9x^2 &= -9\left(x^2 - \frac{4}{3}x - 5\right) \\
&= -9\left[\left(x - \frac{2}{3}\right)^2 - \frac{4}{9} - 5\right] \\
&= -9\left[\left(x - \frac{2}{3}\right)^2 - \frac{49}{9}\right] \\
&= 49 - 9\left(x - \frac{2}{3}\right)^2 \\
&= 49 - (3x-2)^2
\end{aligned}$$

So

$$\int \frac{1}{\sqrt{45 + 12x - 9x^2}}\,\mathrm{d}x = \int \frac{1}{\sqrt{49 - (3x-2)^2}}\,\mathrm{d}x$$

Let

$$u = 3x - 2$$

Then

$$\frac{\mathrm{d}u}{\mathrm{d}x} = 3 \qquad \text{so} \qquad \mathrm{d}x = \frac{1}{3}\,\mathrm{d}u$$

Hence

$$\int \frac{1}{\sqrt{45 + 12x - 9x^2}}\,\mathrm{d}x = \frac{1}{3}\int \frac{1}{\sqrt{49 - u^2}}\,\mathrm{d}u$$

Using the standard result

$$\int \frac{1}{\sqrt{a^2 - x^2}}\,\mathrm{d}x = \arcsin\left(\frac{x}{a}\right) + C$$

with $a = 7$,

$$\begin{aligned}
\int \frac{1}{\sqrt{45 + 12x - 9x^2}}\,\mathrm{d}x &= \frac{1}{3}\arcsin\left(\frac{u}{7}\right) + C \\
&= \frac{1}{3}\arcsin\left(\frac{3x-2}{7}\right) + C
\end{aligned}$$

Hence,

$$\int \frac{1}{\sqrt{45 + 12x - 9x^2}}\,\mathrm{d}x = \frac{1}{3}\arcsin\left(\frac{3x-2}{7}\right) + C$$

13. Without using a calculator, find

(a)

$$\int_{-1/5}^{1} \frac{1}{25x^2 - 20x + 13}\,\mathrm{d}x$$

giving your answer as a multiple of π **[5]**

(b)

$$\int_{-1/5}^{1} \frac{1}{\sqrt{25x^2 - 20x + 13}}\,\mathrm{d}x$$

giving your answer in the form $p\ln(q + r\sqrt{2})$, where p, q and r are rational numbers to be found. **[7]**

Solution

(a) Complete the square in the denominator:

$$25x^2 - 20x + 13 = (5x-2)^2 + 9$$

So

$$\int_{-\frac{1}{5}}^{1} \frac{1}{25x^2 - 20x + 13}\,\mathrm{d}x = \int_{-\frac{1}{5}}^{1} \frac{1}{(5x-2)^2 + 9}\,\mathrm{d}x$$

Let

$$u = \frac{5x-2}{3}$$

Then

$$5x - 2 = 3u, \qquad \mathrm{d}x = \frac{3}{5}\,\mathrm{d}u$$

Change the limits:

$$x = -\frac{1}{5} \;\Rightarrow\; u = \frac{-1-2}{3} = -1, \qquad x = 1 \;\Rightarrow\; u = \frac{5-2}{3} = 1$$

Hence

$$\begin{aligned}\int_{-\frac{1}{5}}^{1} \frac{1}{(5x-2)^2 + 9}\,\mathrm{d}x &= \int_{-1}^{1} \frac{1}{9u^2 + 9} \cdot \frac{3}{5}\,\mathrm{d}u \\ &= \frac{1}{15}\int_{-1}^{1} \frac{1}{1+u^2}\,\mathrm{d}u\end{aligned}$$

Using the standard result

$$\int \frac{1}{1+u^2}\,\mathrm{d}u = \arctan u + c$$

we get

$$\begin{aligned}\frac{1}{15}\int_{-1}^{1} \frac{1}{1+u^2}\,\mathrm{d}u &= \frac{1}{15}\left[\arctan u\right]_{-1}^{1} \\ &= \frac{1}{15}\left(\arctan 1 - \arctan(-1)\right) \\ &= \frac{1}{15}\left(\frac{\pi}{4} - \left(-\frac{\pi}{4}\right)\right) \\ &= \frac{1}{15} \cdot \frac{\pi}{2} \\ &= \frac{\pi}{30}\end{aligned}$$

Therefore, the value of the integral is

$$\frac{\pi}{30}$$

(b) As in part (a),

$$25x^2 - 20x + 13 = (5x-2)^2 + 9$$

So

$$\int_{-\frac{1}{5}}^{1} \frac{1}{\sqrt{25x^2 - 20x + 13}}\,\mathrm{d}x = \int_{-\frac{1}{5}}^{1} \frac{1}{\sqrt{(5x-2)^2 + 9}}\,\mathrm{d}x$$

Let

$$u = \frac{5x-2}{3}$$

Then

$$5x - 2 = 3u, \qquad \mathrm{d}x = \frac{3}{5}\,\mathrm{d}u$$

and the limits again become

$$x = -\frac{1}{5} \Rightarrow u = -1, \qquad x = 1 \Rightarrow u = 1$$

Therefore

$$\begin{aligned}\int_{-\frac{1}{5}}^{1} \frac{1}{\sqrt{(5x-2)^2 + 9}}\,\mathrm{d}x &= \int_{-1}^{1} \frac{1}{\sqrt{9u^2 + 9}} \cdot \frac{3}{5}\,\mathrm{d}u \\ &= \frac{1}{5}\int_{-1}^{1} \frac{1}{\sqrt{u^2+1}}\,\mathrm{d}u\end{aligned}$$

Using the standard result

$$\int \frac{1}{\sqrt{u^2+a^2}}\,\mathrm{d}u = \ln\left|u + \sqrt{u^2+a^2}\right| + c$$

with $a = 1$, we have

$$\int \frac{1}{\sqrt{u^2+1}}\,\mathrm{d}u = \ln\left|u + \sqrt{u^2+1}\right| + c$$

Hence

$$\begin{aligned}\frac{1}{5}\int_{-1}^{1} \frac{1}{\sqrt{u^2+1}}\,\mathrm{d}u &= \frac{1}{5}\left[\ln\left(u + \sqrt{u^2+1}\right)\right]_{-1}^{1} \\ &= \frac{1}{5}\left(\ln(1+\sqrt{2}) - \ln(\sqrt{2}-1)\right) \\ &= \frac{1}{5}\ln\left(\frac{1+\sqrt{2}}{\sqrt{2}-1}\right)\end{aligned}$$

Now simplify the fraction by rationalising:

$$\begin{aligned}\frac{1+\sqrt{2}}{\sqrt{2}-1} &= \frac{(1+\sqrt{2})(\sqrt{2}+1)}{(\sqrt{2}-1)(\sqrt{2}+1)} \\ &= \frac{(1+\sqrt{2})^2}{2-1} \\ &= (1+\sqrt{2})^2 \\ &= 1 + 2\sqrt{2} + 2 \\ &= 3 + 2\sqrt{2}\end{aligned}$$

So the integral is

$$\frac{1}{5}\ln(3 + 2\sqrt{2})$$

Therefore, in the form $p\ln(q + r\sqrt{2})$,

$$p = \frac{1}{5}, \qquad q = 3, \qquad r = 2$$

14. (i) Given that $\tanh y = x$, where $|x| < 1$, show that

$$y = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right)$$

and hence that

$$\operatorname{artanh} x = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right)$$ **[6]**

(ii) Use your result in part (i), or otherwise, to find

$$\int_{1/5}^{2/5} \frac{1}{4-9x^2}\,\mathrm{d}x$$

expressing your answer in an exact logarithmic form. **[6]**

Solution

(i) Starting from the definition of the hyperbolic tangent,

$$\tanh y = \frac{e^y - e^{-y}}{e^y + e^{-y}}$$

Multiplying numerator and denominator by e^y gives

$$\tanh y = \frac{e^{2y} - 1}{e^{2y} + 1}$$

Given that $\tanh y = x$, we have

$$\begin{aligned} x &= \frac{e^{2y} - 1}{e^{2y} + 1} \\ x(e^{2y} + 1) &= e^{2y} - 1 \\ xe^{2y} + x &= e^{2y} - 1 \\ e^{2y} - xe^{2y} &= x + 1 \\ e^{2y}(1 - x) &= 1 + x \end{aligned}$$

Since $|x| < 1$, we have $1 - x > 0$, so we may divide through by $1 - x$:

$$e^{2y} = \frac{1+x}{1-x}$$

Now take natural logarithms:

$$2y = \ln\left(\frac{1+x}{1-x}\right)$$

Hence

$$y = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right)$$

So the required result is shown.

Since $\operatorname{artanh} x$ is the inverse of $\tanh y$, this means

$$\operatorname{artanh} x = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right)$$

(ii) From part (i),

$$\operatorname{artanh} u = \frac{1}{2}\ln\left(\frac{1+u}{1-u}\right)$$

for $|u| < 1$.

Differentiating,

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}u}\left(\operatorname{artanh} u\right) &= \frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}u}\left[\ln(1+u)-\ln(1-u)\right]\\ &= \frac{1}{2}\left(\frac{1}{1+u}+\frac{1}{1-u}\right)\\ &= \frac{1}{2}\left(\frac{(1-u)+(1+u)}{1-u^2}\right)\\ &= \frac{1}{1-u^2}\end{aligned}$$

So

$$\int \frac{1}{1-u^2}\,\mathrm{d}u = \operatorname{artanh} u + c$$

Now let

$$u = \frac{3x}{2}$$

Then

$$x = \frac{2u}{3}, \qquad \mathrm{d}x = \frac{2}{3}\,\mathrm{d}u$$

Also, when $x = \frac{1}{5}$, then $u = \frac{3}{10}$, and when $x = \frac{2}{5}$, then $u = \frac{3}{5}$.
Therefore

$$\begin{aligned}\int_{1/5}^{2/5} \frac{1}{4-9x^2}\,\mathrm{d}x &= \int_{3/10}^{3/5} \frac{1}{4-9\left(\frac{2u}{3}\right)^2}\cdot\frac{2}{3}\,\mathrm{d}u\\ &= \int_{3/10}^{3/5} \frac{1}{4-4u^2}\cdot\frac{2}{3}\,\mathrm{d}u\\ &= \frac{1}{6}\int_{3/10}^{3/5} \frac{1}{1-u^2}\,\mathrm{d}u\\ &= \frac{1}{6}\left[\operatorname{artanh} u\right]_{3/10}^{3/5}\end{aligned}$$

Using the result from part (i),

$$\begin{aligned}\int_{1/5}^{2/5} \frac{1}{4-9x^2}\,\mathrm{d}x &= \frac{1}{6}\left[\frac{1}{2}\ln\left(\frac{1+u}{1-u}\right)\right]_{3/10}^{3/5}\\ &= \frac{1}{12}\left[\ln\left(\frac{1+u}{1-u}\right)\right]_{3/10}^{3/5}\\ &= \frac{1}{12}\left(\ln\left(\frac{1+\frac{3}{5}}{1-\frac{3}{5}}\right)-\ln\left(\frac{1+\frac{3}{10}}{1-\frac{3}{10}}\right)\right)\\ &= \frac{1}{12}\left(\ln 4-\ln\left(\frac{13}{7}\right)\right)\\ &= \frac{1}{12}\ln\left(\frac{28}{13}\right)\end{aligned}$$

Hence the exact value of the integral is

$$\frac{1}{12}\ln\left(\frac{28}{13}\right)$$

15. Show that

$$\int \frac{x^2+14x+21}{(x-1)(x^2+4x+13)}\,\mathrm{d}x = A\ln|x-1| + B\ln(x^2+4x+13) + D\arctan\left(\frac{x+2}{3}\right) + c$$

where A, B and D are constants to be found. **[7]**

Solution

Write

$$\frac{x^2+14x+21}{(x-1)(x^2+4x+13)} = \frac{P}{x-1} + \frac{Qx+R}{x^2+4x+13}$$

Then

$$x^2+14x+21 = P(x^2+4x+13) + (Qx+R)(x-1)$$

Expanding the right-hand side,

$$\begin{aligned} P(x^2+4x+13) + (Qx+R)(x-1) &= Px^2 + 4Px + 13P + Qx^2 - Qx + Rx - R \\ &= (P+Q)x^2 + (4P-Q+R)x + (13P-R) \end{aligned}$$

Comparing coefficients with $x^2+14x+21$ gives

$$\begin{aligned} P+Q &= 1 \\ 4P-Q+R &= 14 \\ 13P-R &= 21 \end{aligned}$$

From the first equation,

$$Q = 1-P$$

From the third equation,

$$R = 13P-21$$

Substitute into the second equation:

$$\begin{aligned} 4P-(1-P)+(13P-21) &= 14 \\ 4P-1+P+13P-21 &= 14 \\ 18P-22 &= 14 \\ 18P &= 36 \\ P &= 2 \end{aligned}$$

So

$$Q = 1-2 = -1, \qquad R = 13(2)-21 = 5$$

Hence

$$\frac{x^2+14x+21}{(x-1)(x^2+4x+13)} = \frac{2}{x-1} + \frac{-x+5}{x^2+4x+13}$$

Therefore

$$\int \frac{x^2+14x+21}{(x-1)(x^2+4x+13)}\,\mathrm{d}x = \int \frac{2}{x-1}\,\mathrm{d}x + \int \frac{-x+5}{x^2+4x+13}\,\mathrm{d}x$$

Now rewrite the second numerator in terms of the derivative of the quadratic:

$$-x+5 = -\frac{1}{2}(2x+4)+7$$

Also,

$$x^2+4x+13 = (x+2)^2+9$$

So

$$\begin{aligned} \int \frac{x^2+14x+21}{(x-1)(x^2+4x+13)}\,\mathrm{d}x &= \int \frac{2}{x-1}\,\mathrm{d}x - \frac{1}{2}\int \frac{2x+4}{x^2+4x+13}\,\mathrm{d}x + 7\int \frac{1}{(x+2)^2+9}\,\mathrm{d}x \\ &= 2\ln|x-1| - \frac{1}{2}\ln(x^2+4x+13) + 7\int \frac{1}{(x+2)^2+3^2}\,\mathrm{d}x \end{aligned}$$

Using the standard result

$$\int \frac{1}{u^2+a^2}\,\mathrm{d}u = \frac{1}{a}\arctan\left(\frac{u}{a}\right) + c$$

with $u = x+2$ and $a = 3$,

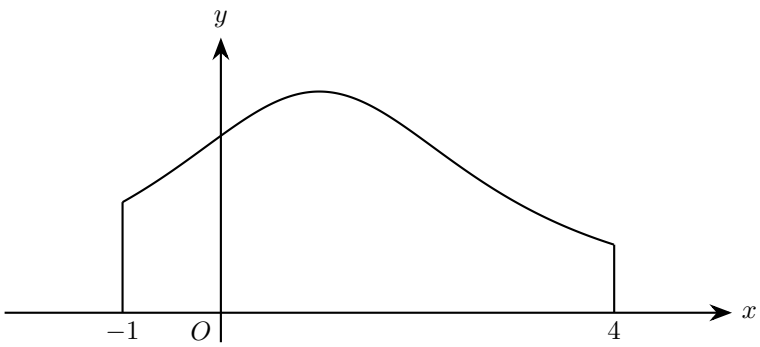
$$\int \frac{1}{(x+2)^2+9}\,\mathrm{d}x = \frac{1}{3}\arctan\left(\frac{x+2}{3}\right)$$

Hence

$$\int \frac{x^2+14x+21}{(x-1)(x^2+4x+13)}\,\mathrm{d}x = 2\ln|x-1| - \frac{1}{2}\ln(x^2+4x+13) + \frac{7}{3}\arctan\left(\frac{x+2}{3}\right) + c$$

So the required constants are

$$A = 2, \qquad B = -\frac{1}{2}, \qquad D = \frac{7}{3}$$



16. The diagram shows the cross-section of a solid wooden beam.
Each unit on the axes represents 6 cm. The curved upper boundary of the beam is modelled by the equation

$$y = \frac{9}{(x-1)^2+4}$$

The cross-section is bounded by the curve, the x-axis and the lines $x = -1$ and $x = 4$.
Assuming that the beam is a prism of length 240 cm, find, correct to 3 significant figures, the volume of the beam. **[6]**

Solution

The cross-sectional area is the area under the curve

$$y = \frac{9}{(x-1)^2+4}$$

from $x = -1$ to $x = 4$.
First find this area in graph units:

$$A_{\text{units}} = \int_{-1}^{4} \frac{9}{(x-1)^2+4}\,\mathrm{d}x$$

To reduce this to a standard form, let

$$u = x - 1 \qquad \Rightarrow \qquad \mathrm{d}u = \mathrm{d}x$$

When $x = -1$, $u = -2$, and when $x = 4$, $u = 3$.
So

$$A_{\text{units}} = \int_{-2}^{3} \frac{9}{u^2+2^2}\,\mathrm{d}u$$

Using the standard result

$$\int \frac{1}{a^2+x^2}\,\mathrm{d}x = \frac{1}{a}\arctan\left(\frac{x}{a}\right) + c$$

with $a = 2$,

$$\begin{aligned}
A_{\text{units}} &= 9\int_{-2}^{3} \frac{1}{u^2+2^2}\,\mathrm{d}u \\
&= 9\left[\frac{1}{2}\arctan\left(\frac{u}{2}\right)\right]_{-2}^{3} \\
&= \frac{9}{2}\left[\arctan\left(\frac{u}{2}\right)\right]_{-2}^{3} \\
&= \frac{9}{2}\left(\arctan\left(\frac{3}{2}\right) - \arctan(-1)\right) \\
&= \frac{9}{2}\left(\arctan\left(\frac{3}{2}\right) + \frac{\pi}{4}\right)
\end{aligned}$$

Now convert to cm^2.

Each unit on each axis represents 6 cm, so

$$1 \text{ square unit} = 6 \times 6 = 36 \text{ cm}^2$$

Hence the cross-sectional area is

$$\begin{aligned} A &= 36A_{\text{units}} \\ &= 36 \cdot \frac{9}{2}\left(\arctan\left(\frac{3}{2}\right) + \frac{\pi}{4}\right) \\ &= 162\left(\arctan\left(\frac{3}{2}\right) + \frac{\pi}{4}\right) \\ &\approx 286.447 \text{ cm}^2 \end{aligned}$$

The beam is a prism of length 240 cm, so

$$\begin{aligned} V &= 240A \\ &= 240 \times 286.447 \\ &\approx 68747.3 \text{ cm}^3 \end{aligned}$$

Therefore, correct to 3 significant figures, the volume of the beam is

$$6.87 \times 10^4 \text{ cm}^3$$

17. Find

$$\int_1^e \frac{1}{x\sqrt{4-(\ln x)^2}}\,\mathrm{d}x$$

expressing your answer in terms of π. **[8]**

Solution

Let

$$u = \ln x$$

Then

$$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{x} \qquad \Rightarrow \qquad \mathrm{d}u = \frac{1}{x}\,\mathrm{d}x$$

We also change the limits:
when $x = 1$, $u = \ln 1 = 0$
when $x = e$, $u = \ln e = 1$
So the integral becomes

$$\int_1^e \frac{1}{x\sqrt{4-(\ln x)^2}}\,\mathrm{d}x = \int_0^1 \frac{1}{\sqrt{4-u^2}}\,\mathrm{d}u$$

Now use the standard result

$$\int \frac{1}{\sqrt{a^2-u^2}}\,\mathrm{d}u = \arcsin\left(\frac{u}{a}\right) + c$$

Here $a = 2$, so

$$\int_0^1 \frac{1}{\sqrt{4-u^2}}\,\mathrm{d}u = \left[\arcsin\left(\frac{u}{2}\right)\right]_0^1$$

Therefore

$$\left[\arcsin\left(\frac{u}{2}\right)\right]_0^1 = \arcsin\left(\frac{1}{2}\right) - \arcsin(0) = \frac{\pi}{6} - 0 = \frac{\pi}{6}$$

Hence the value of the integral is

$$\frac{\pi}{6}$$